

Jyoti Rautela

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RESEARCH VISION

My research explores how technology can be designed WITH people who are often excluded from dominant design processes. Across my projects, I focus on them through three values:

1. By giving them agency and not treating them as passive recipients - Participatory Framework [P1, P2, P4]
2. Surfacing that technology doesn't exist alone, it is embedded in services, relationships and care ecosystem - Embodied Understanding [P1, P2, P3]
3. Using AI to support people's autonomy without taking over interpretation or decision-making - AI that supports [P1, P3]

EDUCATION

IIT Roorkee, Uttarakhand, India July 2024 - May 2026
M.Des. in Industrial Design, DOD CGPA 8.77

NIFT Bhubaneswar, Odisha, India Aug 2016 - Sep 2020
B.Des. in Accessory Design CGPA 8.0

RESEARCH EXPERIENCE

Mobility & Vehicle Design (MVD) Lab Jun- Aug 2025
Research Intern (Advisor: Prof. Unni Mohan M.) IIT Bombay, India

- Helm and cockpit redesign - Kochi Water Metro, Kerala.
| *HCI-Ergonomics-HFD*

Mobility & Vehicle Design (MVD) Lab Sept - Dec 2025
Graduate research Intern (Advisor Prof. Nishant Sharma) IIT Bombay, India

- Research project on a low-floor mobility aid for women with lower-limb impairment, integrating anthropometric data, with co-speculative and participatory design frameworks. | *Disability-Affordability-Built environments-Social infrastructure justice*

TEACHING EXPERIENCE

Department of Design, IIT Roorkee, India; Jan - May 2026
Teaching Assistant (Instructor: Prof. Abinash Kumar Swain)

- **DEC-102 Design Methodology**, B.Des 1st year (35 students) - Introduced speculative and participatory design frameworks, and research-through-embodiment as the third paradigm of HCI; structured rubrics, delivered presentations, and evaluated assignments.
- **DET-101 Solid Modelling**, B.Des 2nd year (35 students) - Led studio sessions on CAD and surface modelling (Rhino, Blender), rendering, engineering drawing, and introductory 3D printing; set rubrics, handled evaluations, exam duties, and after-hours reviews.

PROJECTS

Low floor Assistive Device -

Sept 2025 - May 2026

Understanding Assistive Mobility Barriers for People with Lower Limb Disabilities in domestic Indian context [P1]

- Developed a modular wheelchair rear-wheel prototype
| *Field co-design and participatory study with clinicians & PwDs* (Delhi, Mumbai, Roorkee)
| *Advisor: Prof. Madhan Vasudevan, Prof. Abinash Swain (IITR), Prof. Nishant Sharma (IITB)*

AuraMist - Multimodal Mood Sensing & Just-in-Time Interventions [P2]

Aug - Nov 2024

- Fused wearable skin temperature (digital watch) and eye tracking methods to infer arousal and valence, (paced breathing, focus blocks) with a BLE aroma-haptic device.
| *Advisor: Prof. Saptarishi Kolay (IITR)*

RetroWalk - In-Shoe Gait Imbalance Sensing for Older Adults [P3]

Oct 2019 - Feb 2020

- Built an instrumented insole prototype with on-device fusion and load cells to estimate stride asymmetry and push real-time data via BLE app for safer walking and posture for elderly. | *Advisor: Prof. Harish Rajiv (NIFT Bhubaneswar), Mr. Ankit Sengar (Benetton India Pvt. Ltd.)*

CONFERENCE

Emerge Conference 2026, IIT Bombay

Mar 2026

Full Paper Presentation on Participatory Design for Assistive Mobility in Indian Homes

IDEAS Conference 2024, IIT Roorkee

Oct 2024

Presented Abstract on Tackling Postural Imbalance in Elderly People: A Technological Approach

INDUSTRIAL EXPERIENCE

Myntra Fashion

Mar 2023 - Aug 2024

Associate - Design | Footwear & Accessory

Bangalore, India

- Built 3D CAD models: Sizing analytics-prototype material-aware design collections for Ducati, Nautica, HrX. Trend forecasting, Materials procurement, Sample handoffs.

Rapawalk Pvt. Ltd.

Nov 2021 - Mar 2023

Product development & Sourcing Management | Accessory

Bangalore, India

- Led end-to-end accessory development & sourcing: Material selection, vendor onboarding, QC, and CAD-to-sample handoffs for manufacturable launches.

Benetton India

Dec 2019 - May 2020

Footwear Design Intern

Delhi, India

- Automated footwear product lifecycle: standardized lasts & molds, and worked in R & D department for my final year project [P3]

LEADERSHIP

Department of Design, IIT Roorkee

Sept 2024 - May 2026

Department Placement Representative

- Led department internship/placement ops-employer outreach, recruitment drives, and portfolio clinics-while hosting a “Women in Design” workshop to advance inclusive hiring.

Student Wellness Society, IIT Roorkee

Sept 2024 - May 2025

Women in Design Coordinator (dept-level)

- Led the talks on menstrual health, safety of students in managerial roles - conducted expert sessions/workshops

SKILLS

Methodology: Speculative and Participatory Design, Responsible Innovation, Interdisciplinary Approaches, Inclusive AI systems, Human-Computer Interaction, Research through Design (RtD)

Design & Prototyping Tools: Adobe Creative Cloud, Figma, Unity, Arduino Uno, Rhino

Programming Languages: Python (data processing, prototyping), HTML, CSS, JS.

AWARDS

Silver Medal, **Inter IIT Sports Meet** | *Basketball* | IIT Madras

Dec 2025

Most Valued Player, **Inter IIT Sports Meet** | *Basketball* | IIT Kanpur

Dec 2024

Gold Medal, **Inter IIT Sports Meet** | *Basketball* | IIT Kanpur

Dec 2024

REFERENCE

Dr. Madhan Kumar Vasudevan | *Assistant Professor* | *Thesis Supervisor*

IIT Roorkee, India

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Human-Computer Interaction (HCI), Haptics, Neuroscience, Affective Computing, Machine Learning

Dr. Abinash Kumar Swain | *Associate Professor* | *Thesis Supervisor*

IIT Roorkee, India

abinash.swain@me.iitr.ac.in | +91 7439979147

Mechanical engineering design, HMI, Design Methodology